

# Lab 4 Agent Assignment Conversation

Arjun Srikanth | PES1UG24AM920 | Section H | Assigned game 08\_helicopter

## 1 The student assignment

### Student

Take ownership of Software Engineering Lab 4 for my Section H submission. Work from the assigned starter repository SETAPESU26/08\_helicopter and the supplied README, but first confirm that the assignment maps to SRN PES1UG24AM920 and do not modify my accepted Labs 1-3.

Run the unchanged Pygame game before editing and preserve a genuine ten-second baseline recording. Then implement the four required outcomes in a controlled sequence: cap vertical speed and make Up/Down reversal immediate; contain the complete player at the top and bottom; end the game on contact with either solid wall while keeping the gap safe; add distance scoring with a visible final value and reset-on-restart; and add a visibly active shield that absorbs exactly one collision and turns off immediately after use.

Treat this as an engineering task, not a blind code dump. Inspect the starter, explain the cause of each defect, keep the implementation proportional to the small game, test edge cases, and record a matching after video. Use separate commits for the four tasks, keep credentials and unrelated files out of the submission, and leave me a clean Lab-4 package containing code, both videos, the conversation PDF, and a README link.

### AI Agent

I will first verify the assignment identity and inspect the starter behavior. I will capture the baseline before touching the code, then implement and review Tasks 1-4 in order. Each task will have a focused behavioral check and its own commit so that the history remains easy to evaluate.

I will validate gameplay in the native window as well as through code-level checks: sustained movement, reversal, exact boundary contact, safe gap traversal, lethal wall contact, score progression and reset, one-use shield behavior, restart, and the final ten-second recording. I will keep the final package limited to the requested artifacts and report any ambiguity before expanding scope.

## **2 Iterative implementation**

### **AI Agent**

The assignment is confirmed: the Section H sheet maps PES1UG24AM920 to game 8, 08\_helicopter. The starter keeps the player as its original rounded rectangle and supplies no separate helicopter sprite. I ran the untouched game and retained the before recording so the original behavior remains distinguishable from the fix.

The defects are localized. Movement keeps accumulating velocity without a cap or decisive opposite-direction handling, the lower boundary is missing, collision is not connected to game state, scoring is absent, and the shield requirement is not implemented. I will preserve the existing loop and obstacle model unless a requirement forces a small change.

### **Student**

Keep the changes easy to explain in a viva. At each checkpoint, state what changed, what you tested, and what evidence will appear in the after video. Pay special attention to the complete player rectangle at both boundaries, because a partial containment check would still look broken in gameplay.

### **AI Agent**

Task 1 is complete with bounded vertical speed, immediate direction reversal and release damping. The player is clamped using its full height, so it cannot leave the playfield at either edge. Task 2 now distinguishes gap passage from wall contact: touching a solid wall enters a clear game-over state, while a safe gap remains playable.

Task 3 tracks distance during active play, displays it during the run, preserves the final value after game over, and resets it to zero when a new game starts. Updates stop once the game is over, so the final score is stable and easy to verify.

### **Student**

Implement the shield as a one-use mechanic rather than general invulnerability. I should be able to see when it is active, see it disappear on the first absorbed collision, and observe that a later unprotected collision is lethal. If the first gameplay review is confusing, simplify the behavior instead of adding extra state that is hard to justify.

### **AI Agent**

Space activates the shield and a blue outline makes the active state visible. The shield is consumed on the first collision. During review, I removed unnecessary per-wall contact state and made the protected wall disappear on the absorbed hit; the next solid wall is lethal unless the player activates protection again. This keeps the one-hit contract explicit and avoids a misleading pass-through of an intact wall.

The behavioral suite covers movement, reversal, both boundaries, safe gaps, lethal contact, distance/reset, shield consumption, reactivation and simultaneous contacts. The native after recording is checked for the same sequence: containment, reversal, safe passage, one shield hit, a later game over, and restart.

### **Design decision**

The shield-hit wall is removed as the minimal feedback for an absorbed collision. That is an implementation choice inside the required mechanic; it does not add a new game mode or redesign the starter.

### **3 Review verification and delivery**

#### **Student**

Before handoff, verify the project as a teacher would: confirm the assigned repository, keep accepted Labs 1-3 untouched, check that each task is visible in the implementation, and make sure the videos are genuine ten-second gameplay captures rather than screenshots or synthetic transcripts. Keep the teacher-facing folder simple and make the conversation link obvious.

#### **AI Agent**

The final implementation is scoped to the four assigned tasks. The extended behavioral checks pass, including exact top/bottom contact, long-held keys, safe gap positions, score reset, one-use shield behavior, reactivation, and multiple obstacle contacts. The native after video is ten seconds at 30 fps and is reviewed end to end; the original before video remains unchanged.

The code history keeps Tasks 1-4 separate. The final game shows the original player rendering, bounded movement, wall collision with a readable game-over state, distance during play and at the end, restart with a zeroed score, and a visible shield that is consumed after one hit. No pull request is opened against the instructor repository.

#### **Student**

Use the personal repository only. I want Lab-4 to contain the updated game, before.mp4, after.mp4, one professional conversation PDF, and a concise README that links to the conversation page. Keep state files, raw captures, test logs and intermediate exports outside the teacher-facing folder.

#### **AI Agent**

The package is ready for review. The conversation document is an edited/consolidated duplicate-style account prepared from the verified assignment and project discussion, not a claim of verbatim chat history. The repository presentation remains minimal, the accepted earlier labs are preserved, and the evidence is aligned with the supplied Lab 4 checklist.

#### **Deliverables at a glance**

Updated Pygame source; genuine before gameplay video; genuine after gameplay video; this editable conversation document; and a README link to the conversation page.